

Some Basic Concepts of Chemistry

NCERT Class 11 Chemistry - Chapter 1

1. Importance of Chemistry

Chemistry is often called the central science because it connects physical sciences with life sciences and applied sciences.

It is involved in everything around us — from the food we eat to the medicines we take and the clothes we wear.

Understanding chemistry helps solve global challenges like environmental pollution, climate change, sustainable agriculture, and development of new materials.

- Essential for pharmaceuticals, food processing, and cosmetics
- Vital role in medicine, agriculture, and manufacturing industries
- Helps develop new materials, fuels, and technologies



2. Nature of Matter

Matter is defined as anything that has mass and occupies space. It exists in different states such as solid, liquid, and gas, depending on the arrangement and movement of particles.

Matter can be classified as:

1. *Pure Substances: These have a fixed composition. They are of two types:
 - *Elements: Cannot be broken down into simpler substances (e.g., Hydrogen, Oxygen).
 - Compounds: Formed by chemical combination of elements in fixed ratios (e.g., Water H_2O).
2. *Mixtures: These are physical combinations of two or more substances. They are of two types:
 - *Homogeneous Mixtures: Have uniform composition throughout (e.g., salt in water).
 - Heterogeneous Mixtures: Do not have uniform composition (e.g., sand in water).

- Matter = mass + volume
- Pure substances: elements compounds
- Mixtures: homogeneous and heterogeneous

3. Properties of Matter and their Measurement

Properties of matter are categorized as:

- *Physical Properties: Observable without changing the substance (e.g., color, density, melting point).
- **Chemical Properties: Observed during a chemical change (e.g., flammability, acidity).

Units of Measurement*:

- The International System of Units (SI) is used for standard measurements.
- Some base units include:
 - Length (meter, m)
 - Mass (kilogram, kg)
 - Time (second, s)
 - Temperature (kelvin, K)
 - Amount of substance (mole, mol)
- Physical properties: measurable without changing composition
- Chemical properties: observed during chemical reactions
- SI units ensure consistency in scientific measurements



4. Uncertainty in Measurement

All measurements have some uncertainty or error due to limitations of instruments or observer.

Types of Errors:

- *Systematic Errors: Consistent errors due to faulty equipment or method.

- **Random Errors: Irregular errors due to unknown causes.

Accuracy: How close a measured value is to the true value.

Precision*: How close repeated measurements are to each other.

- No measurement is perfectly accurate
- Systematic error: predictable and consistent
- Random error: unpredictable variations
- Accuracy Precision

5. Significant Figures

Significant figures are the digits in a measurement that are known with certainty plus the first uncertain digit.

Rules:

1. All non-zero digits are significant.
2. Zeros between non-zero digits are significant.
3. Leading zeros are not significant.
4. Trailing zeros in a number with a decimal are significant.

Example: In 0.004506, there are 4 significant figures.

- Used to reflect precision of measurement
- More significant figures = more precise
- Rounding should maintain significant figures

6. Laws of Chemical Combinations

These are the basic principles governing chemical reactions:

1. *Law of Conservation of Mass:

- Mass is neither created nor destroyed in a chemical reaction.

- Total mass of reactants = Total mass of products.

2. **Law of Definite Proportions:**

- A chemical compound always contains the same elements in the same fixed ratio by mass.

3. **Law of Multiple Proportions*:**

- When two elements combine to form more than one compound, the mass of one element combining with a fixed mass of the other bears a simple whole-number ratio.

- Law of Mass Conservation: total mass remains constant
- Law of Definite Proportions: fixed ratio by mass
- Law of Multiple Proportions: whole-number ratios

7. Dalton's Atomic Theory

Proposed by John Dalton in 1808, this theory explained the nature of matter in terms of atoms.

Main postulates:



- All matter is made of indivisible atoms.
 - Atoms of the same element are identical in mass and properties.
 - Atoms combine in simple whole-number ratios to form compounds.
 - Chemical reactions involve rearrangement of atoms; atoms are neither created nor destroyed.
- Matter = atoms
 - Atoms combine in simple ratios
 - Explains laws of chemical combinations

8. Atomic and Molecular Masses

Atomic mass is the mass of one atom of an element, expressed in unified atomic mass unit (u).

1 u = $\frac{1}{12}$ the mass of one atom of carbon-12 isotope.

- **Molecular mass:** Sum of atomic masses of all atoms in a molecule.

- **Formula unit mass:** For ionic compounds, it is the sum of the atomic masses of ions in the formula unit.

Example: Molecular mass of $H_2O = 2 \times 1 + 16 = 18 \text{ u}$

- Atomic mass unit (u) = $1/12$ of C-12 atom
- Molecular mass = total atomic masses in molecule
- Used to calculate moles and stoichiometry

9. Mole Concept and Molar Masses

A *mole* is the amount of substance containing the same number of entities (atoms, molecules, ions) as there are atoms in 12 g of carbon-12.

This number is called *Avogadro's number*: 6.022×10^{23} particles/mol.

- **Molar Mass* is the mass of one mole of a substance. Units: g/mol.

Example: Molar mass of $NaCl = 23 + 35.5 = 58.5 \text{ g/mol}$.

- 1 mole = 6.022×10^{23} particles
- Molar mass = mass of 1 mole in grams
- Links grams to number of particles

10. Percentage Composition

Percentage composition is the percentage by mass of each element in a compound.

Formula:

$$\text{Percentage of element} = \left(\frac{\text{Mass of element in 1 mole of compound}}{\text{Molar mass of compound}} \right) \times 100$$

Example: For water H_2O :

- Mass of H = 2 g, Mass of O = 16 g, Total = 18 g

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- Gives mass
- Helps in finding empirical/molecular formulas

11. Empirical and Molecular Formula

Empirical formula gives the simplest whole-number ratio of atoms of elements in a compound.

Molecular formula gives the actual number of atoms of each element in a molecule.

Steps to find empirical formula:

1. Calculate mass
2. Convert to moles
3. Divide by the smallest number of moles
4. Multiply to get whole numbers

Molecular Formula = $n \times$ (Empirical Formula), where:

$$n = \frac{\text{Molar Mass}}{\text{Empirical Formula Mass}}$$

- Empirical = simplest ratio
- Molecular = actual number of atoms
- Use mass

12. Stoichiometry and Calculations Based on Chemical Equations

Stoichiometry deals with the calculation of reactants and products in a balanced chemical reaction.

Steps:

1. Write the balanced chemical equation
2. Convert known quantities to moles
3. Use mole ratios to find unknown quantities
4. Convert back to required units (mass, volume, particles)

Limiting Reactant: The reactant which is completely consumed first, limiting the amount of product formed.

Theoretical Yield: Maximum product calculated from limiting reactant.

$$\text{Percentage Yield} = \left(\frac{\text{Actual Yield}}{\text{Theoretical Yield}} \right) \times 100$$

- Balanced equations are key to stoichiometry
- Use mole ratio to relate quantities
- Limiting reagent decides the yield

